

Supplementary Material: Proofs for GONS: Self-Calibrated Open-Set Rejection for Few-Shot Class-Incremental Intrusion Detection

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This supplement gives the full assumptions, statements, and proofs of every result in the main paper (Proposition 1, Theorems 1–6, Lemmas 1–4) together with the null-space conditioning analysis. The main paper states each result’s claim in its theory section; nothing here is required to follow the main argument.

1 Proofs

This supplement is self-contained: we prove every numbered result, citing only standard concentration tools: sub-Gaussian quadratic forms and random projections [6], the Hanson–Wright inequality [3], matrix Bernstein [5], the Davis–Kahan variant of [7], the tight DKW constant [2], kernel MMD [1], and the RFF error bound [4]. Absolute constants $C_1, C_2, \dots$ are not tracked tightly; the rates in n, m, r, D, γ are the guaranteed content.

Assumptions.

- (A1) For each known class q , the centred mapped feature $Z_q = \varphi(X) - \mu_q$ ($X \sim P_q$) is sub-Gaussian, $\|Z_q\|_{\psi_2} \leq K$; as RFF coordinates are bounded, $K = O(1)$.
- (A2) The within-class scatter S_w has a rank- r bottom eigenspace ($r = |\mathcal{Q}_t| - 1$) carrying negligible within-class energy, separated from the rest of the spectrum by a gap $\gamma = \lambda_{r+1} - \lambda_r > 0$; $V \in \mathbb{R}^{D \times r}$ ($V^\top V = I_r$) spans it. In the population $\lambda_1 = \dots = \lambda_r = 0$ (an exact null space); in finite samples V is the energy-thresholded bottom eigenspace formed by a ridge-stabilised generalised eigenproblem (see “Null-space existence and conditioning” below).
- (A3) Each known class contributes n i.i.d. support samples to the scatter estimates.
- (A4) The kernel is characteristic, so $P \mapsto \mu_P$ is injective and $\text{MMD}_k(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}}$.
- (A5) Each unknown class u has $\Delta_u = \min_{q \in \mathcal{Q}_t} \text{MMD}_k(P_u, P_q) > 0$.

Proposition 1 (collapse-spread). Under (A2): (a) every known class collapses, $V^\top(\varphi(x_i) - \mu_q) = 0$ and $V^\top \mu_q = s_q$ for any training point x_i of class q (so known training scores are zero); and (b) an unknown class u whose covariance Σ_u is positive on a subspace meeting $\text{range}(V)$ has non-zero projected covariance, $V^\top \Sigma_u V \neq 0$.

Proof. (a) For $v \in \text{range}(V)$, $v^\top S_w v = 0$ gives $\sum_q \pi_q \mathbb{E} \|v^\top(\varphi(X) - \mu_q)\|^2 = 0$; each non-negative summand vanishes, so $v^\top(\varphi(X) - \mu_q) = 0$ a.s. under P_q . (b) If $V^\top \Sigma_u V = 0$ then $\Sigma_u^{1/2} V = 0$, i.e. $\text{range}(V) \subseteq \ker \Sigma_u$, contradicting positivity of Σ_u on a subspace meeting $\text{range}(V)$. $\square$

Theorem 1 (known-score concentration). Under (A1)–(A3), for $x \sim P_q$ and any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$d_*(x)^2 \leq \frac{C_1 K^2 r \log(1/\delta)}{n} + \frac{C_2 K^2 \|S_w\|_{\text{op}}^2 \log(D/\delta)}{\gamma^2 n}.$$

Proof. Write $P = VV^\top$, $\hat{P} = \hat{V}\hat{V}^\top$ for the population and empirical null-space projectors and split $d_q(x) \leq T_1 + T_2 + T_3$ with $T_1 = \|P(\varphi(x) - \mu_q)\|$, $T_2 = \|(\hat{P} - P)(\varphi(x) - \mu_q)\|$, $T_3 = \|\hat{P}(\mu_q - \hat{\mu}_q)\|$ (triangle inequality). (i) $T_1^2 = Z_q^\top P Z_q$ is a sub-Gaussian quadratic form with $\|P\|_F^2 = r$, $\|P\|_{\text{op}} = 1$, and mean $\text{tr}(P\Sigma_q) = 0$ by Proposition 1(a); the quadratic-form bound [6,3] gives $T_1^2 \leq C_1 K^2 r \log(1/\delta)$ w.p. $1 - \delta/3$. (ii) the Davis–Kahan variant [7] gives $\|\hat{P} - P\|_F \leq 2\sqrt{2} \|\hat{S}_w - S_w\|_{\text{op}}/\gamma$, and matrix Bernstein [5] gives $\|\hat{S}_w - S_w\|_{\text{op}} \leq C \|S_w\|_{\text{op}} \sqrt{\log(D/\delta)/n}$; with $\|\varphi(x) - \mu_q\| = O(1)$ (A1), $T_2^2 \leq C_2 K^2 \|S_w\|_{\text{op}}^2 \log(D/\delta)/(\gamma^2 n)$. (iii) $\hat{P}(\hat{\mu}_q - \mu_q)$ sits in $\hat{P}$ ’s rank- r range, so ε -net [6] gives $T_3^2 \leq C_1'' K^2 r \log(1/\delta)/n$. A union bound over the three events with $(T_1 + T_2 + T_3)^2 \leq 3 \sum_i T_i^2$ yields the claim; $d_* \leq d_q$ finishes. $\square$

Lemma 1 (row-space-leakage control). For each known class q , decompose $\mu_u - \mu_q = a_q + b_q$ with $a_q = VV^\top(\mu_u - \mu_q) \in \text{range}(V)$ and b_q in the discarded row space. Writing $\rho_{u,q} = \|b_q\|/\|\mu_u - \mu_q\| \in [0, 1]$ for the leakage of class q and the uniform cap $\rho_u := \max_q \rho_{u,q}$,

$$\begin{aligned} \|V^\top(\mu_u - \mu_q)\| &= \|a_q\| = \sqrt{\|\mu_u - \mu_q\|^2 - \|b_q\|^2} \\ &\geq \text{MMD}_k(P_u, P_q) \sqrt{1 - \rho_{u,q}^2} \geq \Delta_u \sqrt{1 - \rho_u^2} \end{aligned}$$

for every q , using (A5) $\Delta_u = \min_q \text{MMD}_k(P_u, P_q)$.

Proof. Since V is an isometry on its range, $\|V^\top(\mu_u - \mu_q)\| = \|a_q\|$; Pythagoras ($a_q \perp b_q$) gives the middle equality, and the RFF inner product approximates the RKHS norm to $O(D^{-1/2})$. The first inequality is the definition of $\rho_{u,q}$; the second uses $\text{MMD}_k(P_u, P_q) \geq \Delta_u$ and $\rho_{u,q} \leq \rho_u$. The cap ρ_u controls all classes at once: the bound is tight at $\rho_u = 0$ and degrades as $\rho_u \rightarrow 1$. $\square$

Theorem 2 (unknown-score separation). Under (A1)–(A5), for $x \sim P_u$ write the projected unknown score as $d_*(x) \geq \tilde{\Delta}_u - E_n - R(x)$, where $\tilde{\Delta}_u := \Delta_u \sqrt{1 - \rho_u^2}$ is the null-space-projected separation (Lemma 1, A5), E_n an *estimation* term, and $R(x)$ a *per-flow* residual. Then (a) location: the mean score obeys $\mathbb{E}_{x \sim P_u}[d_*(x)] \geq \tilde{\Delta}_u - E_n - C_4 O(D^{-1/2} \sqrt{\log(1/\delta)})$, where $E_n \leq C_3 K \sqrt{\log(1/\delta)/n}$ collects the null-space-projector and signature estimation error and *vanishes as the known-class support n grows*; and (b) per-flow tail: the residual $R(x) := \tilde{\Delta}_u - E_n - d_*(x)$ is sub-Gaussian with proxy σ (the unknown-score parameter, *independent of n*), so $\Pr_{x \sim P_u}[d_*(x) \leq \tilde{\Delta}_u - E_n - t] \leq \exp(-t^2/2\sigma^2)$ for any $t \geq 0$. *Proof.* Fix any known class q . The reverse triangle inequality in the projected space gives $d_q(x) \geq \|V^\top(\mu_u - \mu_q)\| - \|V^\top(\varphi(x) - \mu_u)\|$, and by Lemma 1 $\|V^\top(\mu_u - \mu_q)\| \geq \tilde{\Delta}_u - O(\varepsilon_{\text{RFF}})$ for every q , so the bound passes to the minimum: $d_*(x) \geq \tilde{\Delta}_u - \|V^\top(\varphi(x) - \mu_u)\| - O(\varepsilon_{\text{RFF}})$. Split the projector $V = V_{\text{pop}} + (\hat{V} - V_{\text{pop}})$. The estimated-projector part of $\|V^\top(\varphi(x) - \mu_u)\|$ is at most $\|\hat{V} - V_{\text{pop}}\|_{\text{op}} \|\varphi(x) - \mu_u\| \leq C_3 K \sqrt{\log(1/\delta)/n}$ by Davis–Kahan [7] and matrix Bernstein [5] as in Theorem 1(ii); this is E_n , which vanishes in n . The remaining population term $V_{\text{pop}}^\top(\varphi(x) - \mu_u)$ has mean zero under P_u and, since V_{pop} does not annihilate $\text{Cov}(\varphi \mid P_u)$ (Proposition 1(b)), is a mean-zero sub-Gaussian quadratic form with parameter σ *independent of n* ; this is $R(x)$, giving the one-sided tail in (b). The RFF–RKHS error $\varepsilon_{\text{RFF}} = O(D^{-1/2} \sqrt{\log(1/\delta)})$ [4] is the last term. $\square$

Theorem 3 (finite-sample, distribution-free threshold). Let $\tau = \hat{F}_{\text{known}}^{-1}(1 - \alpha)$ be the empirical $(1 - \alpha)$ -quantile of m known calibration scores. With probability at least $1 - \delta$,

- (i) calibration admission rate $\geq 1 - \alpha$;
- (ii) test admission rate $\geq 1 - \alpha - \sqrt{\log(2/\delta)/2m}$;
- (iii) $\Pr_{X \sim P_u}[d_*(X) > \tau] \geq 1 - \exp(-(\tilde{\Delta}_u - E_n - \tau)^2/2\sigma^2)$
when $\tilde{\Delta}_u - E_n > \tau$.

Here σ^2 is the sub-Gaussian parameter of the per-flow residual $R(X)$ of Theorem 2 (independent of n), E_n its estimation term (vanishing as n grows), and m in (ii) is the *known*-calibration size. Part (iii) is a *per-flow* probability, independent of m ; the empirical TUR over m_u unknown test flows concentrates around it by Hoeffding (an extra m_u factor). The admission rate $1 - \text{FRR}_{\text{known}}$ upper-bounds closed-set accuracy ($\text{CCR} \leq \text{admission rate}$): an accepted known still needs the nearest-signature head correct, a gap outside DKW’s scope.

Proof. Part (i) holds by definition of the empirical quantile. Part (ii) is the Dvoretzky–Kiefer–Wolfowitz inequality with Massart’s tight constant, which is distribution-free and so transfers the calibration admission rate to the test distribution up to an $O(1/\sqrt{m})$ term, uniformly in τ . Part (iii) writes $d_*(X) \geq \tilde{\Delta}_u - E_n - R(X)$ with R the sub-Gaussian unknown residual (Theorem 2), so $\Pr[d_*(X) \leq \tau] \leq \Pr[R(X) \geq \tilde{\Delta}_u - E_n - \tau] \leq \exp(-(\tilde{\Delta}_u - E_n - \tau)^2/2\sigma^2)$ by the

one-sided sub-Gaussian tail (valid since $\tilde{\Delta}_u - E_n > \tau$), per flow and free of m . $\square$

Lemma 2 (Boole K -floor: no closed-form distribution-free TUR certificate). Fix a session with K known classes and acceptance region $A = \bigcup_s B(s, \delta_s)$, and suppose every known signature's ball is no smaller than the unknown's projected spread, $\delta_s^2 \geq \lambda_{\max}(V^\top \Sigma_u V)$ (the “loose-ball” regime). Then every uniform-weight per-signature (or per-class-deduplicated) union bound on $\Pr_{X \sim P_u}[X \in A]$ is at least $K \geq 2 > 1$, hence uninformative as a rejection certificate.

Proof. The per-set anti-concentration factor for a loose set is the empty product $\prod_{j: \lambda_j > \delta_s^2}(\cdot) = 1$; a summing bound contributes one such unit term per loose class, so the total is $\geq K_{\text{loose}} = K$. $\square$

We confirm this K -floor across 48 configurations and check that the non-summing routes (concentration, anti-concentration, low-dimensional tube, radius-cap) are uninformative or invalid at the operating point; hence Theorem 3(iii) is the strongest closed-form, distribution-free statement for the unknown rate. The no-go is specific to *closed-form, distribution-free, uniform-weight* certificates, not to data-dependent or distributional bounds.

Theorem 4 (Composition). Combining Theorems 1–3, whose four concentration events (known concentration, unknown separation, and the DKW and Hoeffding tails of Theorem 3) each hold at confidence $1 - \delta/4$, a union bound yields all four properties simultaneously with probability at least $1 - \delta$. $\square$

Lemma 3 (deployed grid-search reduces to the quantile bound). In deployment τ is the largest grid point ($|\mathcal{G}| \leq 64$) meeting an empirical coverage floor $\hat{F}_{\text{known}}(\tau) \geq \beta$ and a false-known cap (with a 0.95-quantile fallback when infeasible). Then $\hat{\tau}_{\text{dep}}$ inherits the Theorem 3 admission bound at level $1 - \beta$ with *no* $|\mathcal{G}|$ union term. *Proof.* The feasibility check returns $\hat{\tau}_{\text{dep}}$ only if the empirical coverage $\hat{F}_{\text{known}}(\hat{\tau}_{\text{dep}}) \geq \beta$, so this holds at the returned point *by construction* (no grid-resolution caveat). The DKW event of Theorem 3(ii) is $\sup_t$ -uniform, hence $F_{\text{known}}(\hat{\tau}_{\text{dep}}) \geq \hat{F}_{\text{known}}(\hat{\tau}_{\text{dep}}) - \varepsilon \geq \beta - \varepsilon$ even though $\hat{\tau}_{\text{dep}}$ is data-dependent (selected over negative scores); there is nothing to union over, so no $|\mathcal{G}|$ term enters the admission side. When the feasible set is empty the rule returns the 0.95-quantile, the single Theorem 3 object at $\alpha = 0.05$, giving $\geq 0.95 - \varepsilon$. $\square$

Theorem 5 (cross-session admission; unconditional). Across T consolidated refreshes, with probability $\geq 1 - \delta$ and simultaneously for all $t \leq T$, the known-class admission rate stays $\geq \beta - \Theta(\sqrt{\log T/m_t})$, accumulating only the DKW slack, with no subspace-drift term (every retained class is re-projected through the current V_t) and no grid term (Lemma 3); the bound is unconditional and survives $\gamma_{\min} \rightarrow 0$. *Proof.* Union-bound the per-session DKW events of Theorem 3(ii) over the $T+1$ refreshes ($\delta/(T+1)$ each), which inflates each radius by a $\sqrt{\log T}$

factor. The consolidated refresh re-projects every retained class through the current V_t , so the threshold and each store-resident class's score share the session- t geometry and no drift term enters; Lemma 3 removes the grid term. $\square$

Remark (cross-session rejection, partial). The rejection margin $\eta_t = \tilde{\Delta}_u - \tau_t$ (with $\tilde{\Delta}_u = \Delta_u \sqrt{1 - \rho_{\max}^2}$ under a uniform leakage cap $\rho_u \leq \rho_{\max} < 1$) is *not* controlled unconditionally. Telescoping the per-refresh Davis–Kahan rotations gives $\eta_T \geq \eta_0 - (2\sqrt{2}L/\gamma_{\min}) \sum_{t=1}^T \|\Delta \hat{S}_t\|_{\text{op}}$ with $\Delta \hat{S}_t = \hat{S}_{w,t} - \hat{S}_{w,t-1}$, but the slope is $\Theta(1/\gamma_{\min})$ with an unquantified Lipschitz constant L (and $\gamma_{\min}$ is a base-class-separation proxy, not the literal eigen-gap), so we claim *no* closed cross-session rejection certificate. We instead report the per-session rejection rates directly (Section 7.4 of the main paper), consistent with the no-go of Lemma 2.

Theorem 6 (null-space is the uniquely concentrating projection). Fix rank r and let $\kappa(W) = \|W^\top S_w W\|_{\text{op}}$ be the within-class energy retained by a rank- r W . Then

$$\kappa(V_{\text{null}}) = 0, \quad \kappa(V_{\text{PCA}}) \geq \lambda_{r+1}(S_w) \cos^2 \theta > 0, \quad \kappa(V_{\text{rand}}) = \Theta(\bar{\lambda}_w r/D) \text{ w.h.p.,}$$

and empirically $\kappa(\hat{V}_{\text{null}}) \leq C \|S_w\|_{\text{op}} \sqrt{\log(D/\delta)/n} \rightarrow 0$.

where θ is the largest principal angle between $\text{range}(V_{\text{PCA}})$ and $\text{range}(V_{\text{null}})$, V_{rand} is Haar-random, and $\bar{\lambda}_w = \text{tr}(S_w)/D$. *Proof.* For V_{null} , $V^\top S_w V = 0$ by definition, and the empirical bound is matrix Bernstein [5] as in Theorem 1(ii). For V_{PCA} , Courant–Fischer gives $\max_{v \in \text{range}(V_{\text{PCA}}), \|v\|=1} v^\top S_w v \geq \lambda_{r+1}(S_w) \cos^2 \theta$, positive whenever PCA loads any row-space direction. For V_{rand} , $\mathbb{E} v^\top S_w v = \bar{\lambda}_w$ for a uniform unit v and the operator norm concentrates at $\Theta(\bar{\lambda}_w r/D)$ by the random-projection bound [6]. Hence only the null space drives the Theorem 1 constant to zero. $\square$

Lemma 4 (min-distance beats an energy threshold in calibration variance). Let $s_{\text{en}}(x) = -T \log \sum_q e^{-d_q(x)^2/T}$, $s_{\text{min}}(x) = \min_q d_q(x)^2$, with boundary mass $\pi_b(\alpha) = \Pr[d_{(2)}^2 - d_{(1)}^2 < T]$. Then $\text{Var}(\hat{\tau}_\alpha^{\text{en}}) \geq T^2 \log(1 + (|\mathcal{Q}_t| - 1)) \pi_b(\alpha)$, while $\text{Var}(\hat{\tau}_\alpha^{\text{min}}) = O(\sigma_{\text{cal}}^2/m)$ (DKW), free of T and of boundary mass. *Proof.* Point-wise $s_{\text{en}} = s_{\text{min}} - G_T$ with smoothing offset $G_T(x) = T \log(1 + \sum_{q \neq (1)} e^{-\delta_q/T}) \in [0, T \log |\mathcal{Q}_t|]$, which spikes for boundary points ($\delta_{(2)} < T \Rightarrow G_T \geq T \log(1 + e^{-1})$). The empirical-quantile variance is $\alpha(1 - \alpha)/(mf^2)$; the bimodal offset thins the density of s_{en} at the threshold, contributing at least the squared offset times the boundary mass, giving the bound. s_{min} carries no offset, so its quantile obeys the DKW variance of Theorem 3. $\square$

Null-space existence and conditioning (A2 in practice). Because $D = 512$ exceeds the active support rank, S_w is rank-deficient and a genuine bottom-energy subspace exists; GONS needs no exact zero eigenvalues. It takes V as the bottom eigenvectors of (S_w, S_t) by an energy threshold (floored to a minimum dimension,

capped at $n-2$), stabilised by a ridge $S_t + \rho I$ with ρ escalated on ill-conditioning and a deterministic Nyström fallback if the gap is degenerate. So A2’s gap is numerical, not exact, and well-defined even when S_w is near-full-rank.

2 Robustness remarks

Calibration self-seeding (just-admitted class). A class admitted at session t has no held-out data, so its $K=5$ support flows are scored as its first calibration entries. By Proposition 1(a) these training points collapse to 0 under the null space, so they fall below τ *by construction* (coverage 1.0 for that class). Their effect on the *pooled* empirical $(1 - \alpha)$ -quantile is bounded by their share of the calibration pool: reusing K entries among m_t pooled scores moves the quantile by at most $\approx K/m_t$ in rank, which $\rightarrow 0$ as m_t grows ($m_t \geq 5|Q|$ even at its smallest). The admission-rate inflation attributable to one just-admitted class is therefore at most $\approx K/m_t$ and vanishes as held-out data accrues.

Few-shot label noise. Suppose one of the $K=5$ support flows of a class q is mislabelled. Its signature $s_q = V^\top \mu_q$ uses the empirical mean of K points, so a single corrupted flow shifts s_q by at most $\|V^\top(\varphi_{\text{wrong}} - \varphi_{\text{true}})\|/K$. The admission bound (Theorems 3, 5) is computed on the *established-class held-out* calibration pool, not on the few-shot support labels, so it is unaffected by support mislabelling; the sensitivity is confined to the just-admitted class’s own signature (and decreases as that class accrues held-out data).

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